Project Development Phase
Model Performance Test

Date	01.04.2026
Team ID	69bf8031c1eab70902258949
Project Name	Enforcing Mandatory Fiedls Using UI Policies and Migrating Changes with Update Sets
Maximum Marks	

1. Machine Learning for UI Policies

Standard UI Policies rely on hardcoded conditions (e.g., `If Category is Hardware, then Serial Number is mandatory`). ML can make these dynamic.

Dynamic Mandatory Enforcement

- **Predictive Field Requirements:** Instead of a massive list of `if/else` statements, an ML model (Classification) can analyze the "Short Description" and "Description" of a record. If the model predicts a high probability that an incident is "Network" related, it can trigger a UI Policy to make "Router ID" mandatory, even if the user hasn't selected a category yet.
- **Data Integrity Prediction:** You can use ML to detect "form-filling fatigue." If an agent consistently enters placeholder data (like "NA" or "test") to bypass mandatory fields, a model can flag the record for review or prevent submission until "meaningful" data is provided.

Implementation Logic

- **Train:** Use **ServiceNow Predictive Intelligence** to train a model on historical high-quality records.
- **API Call:** Use an **onChange** Client Script that calls an asynchronous GlideAjax.
- **Action:** The script sends the current field values to the ML model, receives a prediction, and uses `g_form.setMandatory()` to enforce fields based on the result.

2. ML for Migrating Changes with Update Sets

Managing Update Sets is often the most manual part of a ServiceNow Admin's job. ML can reduce "collision" risks and migration failures.

Conflict & Collision Prediction

- **Collision Detection:** Before moving an Update Set to Test or Production, an ML model can analyze the XML payloads against the target environment's current state. It can predict the likelihood of a **commit error** based on historical migration failures (e.g., "This Update Set modifies a UI Policy that was changed in Production yesterday").
- **Automated Grouping:** ML can suggest "Related Updates." If you forget to include a specific Script Include or System Property in your Update Set, a clustering model can

identify that these components are typically migrated together and alert you before you close the set.

Semantic Search for Update Sets

- Instead of searching for Update Sets by name, ML-powered semantic search allows you to find logic. You could search for "Mandatory field logic for hardware" to find all related Update Sets across different developers, reducing duplicated effort.

3. Workflow Comparison: Static vs. ML-Enhanced

Feature	Static Approach (Standard)	ML-Enhanced Approach
UI Policy Trigger	Rigid conditions (State = X).	Predictive (Context suggests Category = Y).
Mandatory Fields	Always the same for a category.	Variable based on the complexity of the description.
Update Sets	Manual peer review for conflicts.	Automated risk scoring for migrations.
Error Handling	Fail on commit in target instance.	Predicted failure during development.